

Toh Yan Hui

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EDUCATION

National University of Singapore

Aug 2024 – May 2028

Bachelor of Computing (Honours) in Computer Science

- **CAP:** 4.74/5.00 (First Class Honours), Ranked among the Top 10 students in a cohort of 611 for CS2103T Software Engineering
- **Intended specialisations:** Software Engineering, Artificial Intelligence

Lund University

Aug 2026 – Jan 2027

Exchange Student, Faculty of Engineering (LTH)

- Coursework: EDAN55 Advanced Algorithms, FRTN90 Cloud Computing, EDAN20 Language Technology, EDAF80 Computer Graphics

WORK EXPERIENCE

HMI Managed Healthcare

May 2026 – Aug 2026

AI & Software Engineering Intern

- Improved AI-assisted claim adjudication accuracy from ~30% to ~90% across more than 10,000 historical claims.
- Built and deployed a FastAPI-based OCR PII redaction service on NVIDIA DGX Spark, redacting names and NRICs in ~0.1 seconds per page.
- Implemented and evaluated four end-to-end OCR/PII redaction pipelines, selecting PP-OCRV6 Tiny for its balance of accuracy, text localisation, latency, and on-premises deployability.
- Integrated the redaction service into the existing claims workflow, containerising it with Docker and building PaddlePaddle from source for DGX Spark deployment.

National University of Singapore, School of Computing

May 2026 – Aug 2026

Teaching Assistant, CP2106 Independent Software Development Project (Orbital)

- Served as an Adviser to 13 student teams (26 students), guiding project progress, milestone planning, and overall direction throughout the development cycle.
- Conducted monthly milestone evaluations and provided detailed, actionable feedback to help teams identify gaps, overcome roadblocks, and iteratively improve their projects.

National University of Singapore, School of Computing

Jan 2026 – Apr 2026

Teaching Assistant, CS2103T Software Engineering

- Conducted weekly Software Engineering tutorials covering the Software Development Life Cycle (SDLC), software engineering principles including SOLID, as well as UML, software architecture, design patterns, automated testing, CI/CD, and code review.

PROJECTS

TokTok – Automated Online Review Classifier (TikTok TechJam 2025)

Aug 2025 – Aug 2025

Tech Stack: Python, PyTorch, Hugging Face, pandas, scikit-learn, Streamlit, Google Colab

- Developed an automated review classifier with 4 teammates to detect Advertisement, Irrelevant Content, and Rant Without Visit in online reviews.
- Fine-tuned a DeBERTa model for sequence classification using a preprocessing pipeline, LLM-assisted labelling, and rule-based checks.
- Addressed class imbalance using oversampling and focal loss, and built a Streamlit application for interactive review classification and result visualisation.

LinkedPair – AI-Powered Matchmaking App

May 2025 – Aug 2025

Tech Stack: React Native, JavaScript, Firebase, OpenAI API, Expo

- Developed a cross-platform mobile app with a teammate featuring AI-powered buddy and romantic matching, geolocation-based matching, advanced filters, and chat.
- Integrated Firebase Authentication, Firestore, and Storage, and used the OpenAI API to generate personalised profile descriptions and AI compatibility scores.

Momo – Task Manager with GUI Optimised for CLI

Aug 2025 – Sep 2025

Tech Stack: Java, JavaFX, CSS, Gradle

- Developed a Java task manager for to-dos, deadlines, and events, supporting persistent storage, task search and status updates, with colour-coded feedback and built-in help.

TECHNICAL SKILLS

Programming Languages: Python, Java, TypeScript, JavaScript, C, C++, SQL

AI/ML: PyTorch, Hugging Face, scikit-learn, NumPy, pandas

Frameworks & Technologies: HTML, CSS, React Native, Streamlit, Docker, Firebase, MongoDB, PostgreSQL, Microsoft Azure